DESIGN AND DEVELOPMENT OF IOT BASED EYE CARE PROTECTION SYSTEM

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Abstract *– In today's digital world, televisions, smartphones, and other digital devices have become an essential part of daily life. Many people spend several hours watching TV for entertainment, education, or information. However, sitting too close to a television screen for long periods can cause eye strain, fatigue, headaches, and other vision-related problems. Children are especially at risk, as they often sit very close to the screen. Therefore, maintaining a safe viewing distance is important for protecting eye health and promoting healthy screen habits.This project proposes an Internet of Things (IoT)-based system that monitors the distance between the viewer and the TV screen in real time. The system uses distance-measuring sensors, such as ultrasonic or infrared sensors, to continuously detect how far a person is sitting from the screen and send the data to a microcontroller or IoT device. The microcontroller processes the data and compares it with a predefined safe distance.If the viewer moves closer than the recommended distance, the system generates a warning alert through an LED, sound, or on-screen message, reminding the user to move back. If the warning is ignored for a few seconds, the system activates a second safety mechanism by automatically turning off the TV to prevent prolonged exposure at an unsafe distance.Once the viewer moves back to a safe distance, the system detects the change and allows the TV to turn on again. The system can also analyze usage data and present it as charts or reports, helping users understand their viewing habits. A mobile app can provide recommendations such as maintaining proper distance, taking breaks, and limiting screen time. Parents can monitor their children's TV usage and ensure healthy practices.*

Keywords*: Arduino Uno, GSM SIM800C Module, Ultrasonic Sensor, IR Sensor, Relay Module, Switch, Connecting Cables, 12V Adapter, IoT-Based Eye Protection System, Automatic Screen Control.*

I. Introduction

In today's digital world, televisions and electronic screens have become an essential part of everyday life. People spend many hours watching TV for entertainment,education, and information. However, sitting too close to the television screen for a long period can lead to eye strain, headaches, blurred vision, and other longterm eye problems. Children are especially at risk because they often sit very near to the screen without realizing the harmful effects. Maintaining a safe viewing distance is

important to protect eye health and prevent vision-related issues. Most televisions do not have a built-in system to monitor the viewer's distance. Parents or elders may not always be able to supervise continuously. To solve this problem, an automated and intelligent system is required that can monitor the viewing distance in real time and take necessary action when the distance becomes unsafe.

This project presents an IoT-Based Smart TV Eye Protection System designed to ensure safe television viewing habits. The system uses an Ultrasonic Sensor to measure the distance between the viewer and the TV screen and an IR sensor to detect human presence. An Arduino Uno acts as the main controller to process the sensor data. If the viewer sits too close, the system first provides a warning alert. If the unsafe distance continues for a certain time, the relay module automatically turns off the TV to protect the eyes. Additionally, the GSM SIM800C module can send notification messages to inform parents or users

II. EASE OF USE

A. Easy installation

The proposed IoT-Based Smart TV Eye Protection System is designed for simple and quick installation. The hardware components such as the Ultrasonic Sensor, IR Sensor, Arduino Uno, Relay Module, and GSM SIM800C module are easily available and can be connected using basic wiring techniques. The sensors are placed near the television to detect viewer presence and measure distance accurately. The system is powered using a 12V adapter, and once all components are connected properly, it begins functioning automatically. No complex configuration or advanced technical knowledge is required for setup, making it suitable for home use.

B. User-Friendly Operation

The system operates automatically without requiring continuous monitoring. When a viewer sits too close to the TV, the system first provides a warning alert. If the unsafe distance continues, the relay module turns off the television automatically to protect the viewer's eyes. Once the viewer moves back to a safe distance, normal operation resumes. Additionally, the GSM module can send alert messages to parents or users for extra safety. Since the system works in real time and requires minimal human intervention, it ensures convenience and promotes healthy viewing habits effectively.

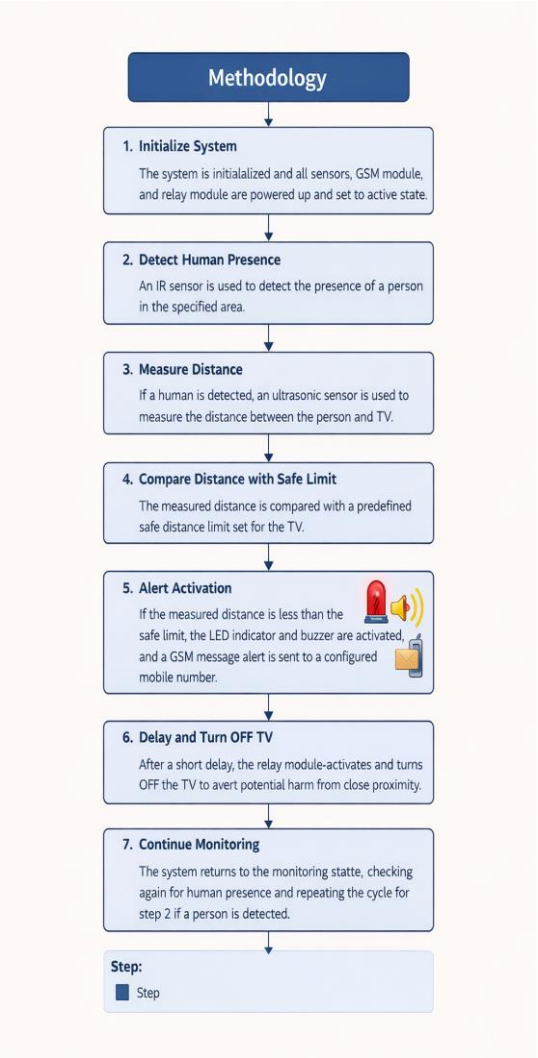
III. Methodology

The methodology of the The methodology of the proposed system describes the step-by-step process used tomeasurethedistance of an object using an ultrasonic sensor and provide visual feedback throughLEDs. Theprocess begins with system initialization, where the Arduino microcontroller and the HC-SR04ultrasonic sensor are configured. During this stage, the Arduino sets the required input andoutputpins, including the trigger pin, echo pin, and LED pins.This ensures that all hardware componentsare The process begins with **system** initialization,where the Arduino microcontroller and all components such as the HC-SR04 ultrasonic sensor, IR sensor, LEDs, buzzer, relay module, and GSM module are configured. The Arduino sets the required input and output pins, including the trigger pin, echo pin, and indicator pins, ensuring that all hardware components are ready for operation.

Next, the IR sensor detects human presence in front of the television. If no person is detected, the system remains in an idle state to save power. Once a person is detected, the system proceeds to measure the distance.

The ultrasonic sensor measures the distance by sending ultrasonic waves through the trigger pin and receiving the reflected signal through the echo pin. The Arduino calculates the distance based on the time taken for the echo to return.

The calculated distance is then compared with a predefined safe distance (for example, 1.5 meters). Based on this comparison:



III.1 Project Architecture

The project architecture of the IoT-Based Smart TV Eye Protection System defines the structure and interaction of all hardware components used to monitor and control the viewing distance. It is designed in a layered approach consisting of input, processing, and output units that work together to ensure safe television usage.

At the input layer, sensors are responsible for collecting real-time data from the environment. The IR sensor detects the presence of a viewer in front of the television, ensuring that the system operates only when a person is present. The ultrasonic sensor (HC-SR04) continuously measures the distance between the viewer and the TV screen by transmitting ultrasonic waves and receiving the reflected signals. This distance data is essential for determining whether the viewer is maintaining a safe position.

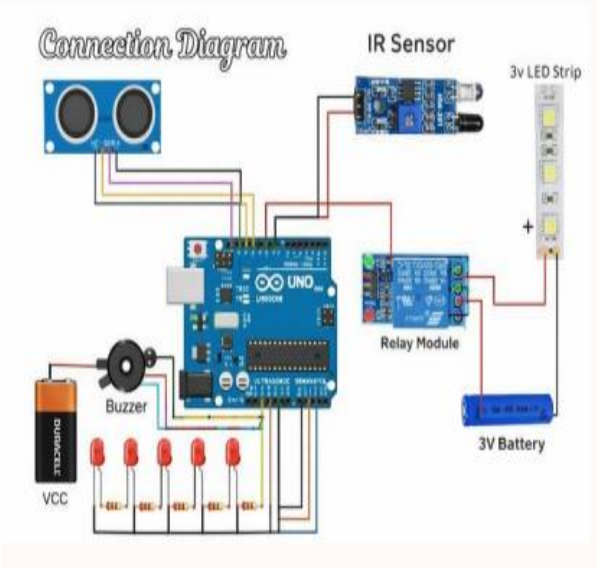
The processing layer consists of the Arduino Uno microcontroller, which acts as the brain of the system. It receives input signals from both sensors and processes the data. The Arduino calculates the distance using the time taken for the ultrasonic signal to return and compares it with a predefined safe distance value. Based on this comparison, the controller makes decisions regarding the system's response.

The decision-making mechanism is an important part of the architecture. If the measured distance is within the safe range, the system continues normal operation without any interruption. However, if the viewer is detected to be too close to the screen, the system initiates a warning phase. If the unsafe condition persists for a specific duration, the system escalates the response by triggering automatic control actions.

The output layer includes devices that interact with the user and control the television. LEDs and a buzzer are used to provide visual and audio alerts, informing the viewer to move back. If the warning is ignored, the relay module is activated to cut off the power supply to the TV, thereby protecting the viewer's eyes. Additionally, the GSM SIM800C module is integrated to send notification messages to parents or users, enabling remote awareness and monitoring.

The architecture follows a continuous monitoring loop, where sensor data is constantly updated and processed in real time.

This ensures immediate detection of unsafe conditions and quick response from the system. The design is simple, cost-effective, and easily scalable for future enhancements such as mobile app integration, cloud data storage.



III.2 Data Flow Diagram

Data flow procedure step-by-step

The flowchart represents the working process of the IoT-Based Smart TV Eye Protection System, showing how the system monitors the viewer's distance and takes action to ensure eye safety.The process begins with the **Start** step, where the system is powered on. In the next step, the system performs initialization, where all components such as sensors (IR and ultrasonic), GSM module, and relay are configured. This ensures that the system is ready to operate properly.After initialization, the system checks for human presence using the IR sensor. This step is important to avoid unnecessary operation when no one is in front of the TV.

- If no person is detected, the system continues monitoring in a loop until someone is present.
- If a person is detected, the system proceeds to the next step.

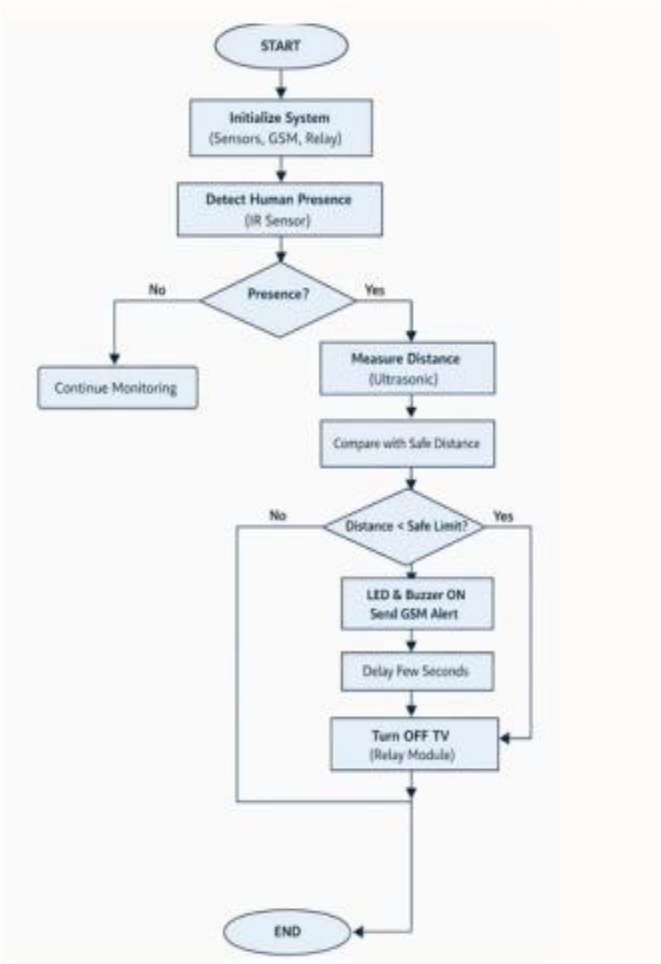
Once presence is confirmed, the system uses the ultrasonic sensor to measure the distance between the viewer and the TV screen. The measured distance is then compared with a predefined safe distance limit.

Based on this comparison:

- If the distance is within the safe limit, the system continues monitoring without any action.
- If the distance is less than the safe limit (viewer too close), the system activates a warning mechanism.

In the warning stage, the LED and buzzer are turned ON, and a GSM alert message may also be sent to notify the user or parents. This acts as the first level of protection, encouraging the viewer to move back.

If the viewer does not respond to the warning and continues to stay too close for a few seconds (delay period), the system activates the relay module to turn OFF the TV. This ensures that prolonged exposure at an unsafe distance is prevented.



IV. Implementation & Experimental Setup

The proposed IoT-Based Smart TV Eye Protection System is implemented using a combination of sensors, a microcontroller, and output devices. The core of the system is the Arduino Uno, which acts as the central processing unit. All components are connected to the Arduino according to the circuit design.

The ultrasonic sensor (HC-SR04) is used to measure the distance between the viewer and the TV screen. It is connected to the Arduino through trigger and echo pins. The **IR sensor** is used to detect human presence and is connected to a digital input pin of the Arduino.

The LEDs and buzzer are connected to output pins and are used to provide visual and audio alerts when the viewer is too close to the screen. The **relay module** is connected to control the power supply of the television. When activated, it turns OFF the TV automatically.

Additionally, the GSM SIM800C module is interfaced with the Arduino using serial communication (TX and RX pins). This module is used to send SMS notifications to users or parents when unsafe viewing conditions are detected.

The system is programmed using the Arduino IDE. The code continuously reads sensor data, processes it, and performs actions based on predefined conditions such as safe distance and delay time

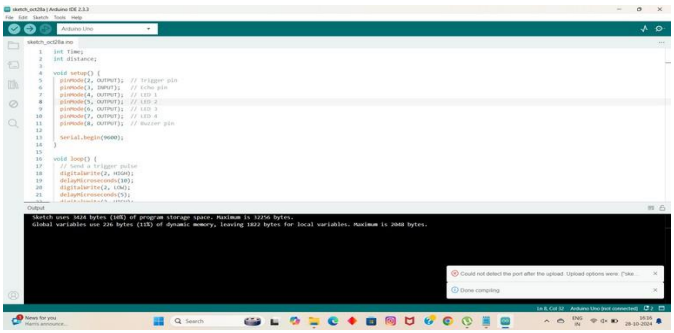


Fig.1: Arduino Program Code

This image shows The experimental setup consists of placing the system in front of a television in a controlled indoor environment. The ultrasonic sensor is positioned in such a way that it directly faces the viewer to accurately measure the distance. The IR sensor is aligned to detect human presence within the viewing area.

The relay module is connected to the TV's power supply, allowing the system to control the ON/OFF state of the television. LEDs and buzzer are placed in visible and audible positions for effective alerts.

A predefined safe distance (for example, 1.5 meters) is set in the system. Different test cases are performed by varying the distance of the viewer from the TV:

- When the viewer is at a safe distance, the system remains inactive.
- When the viewer moves closer than the safe limit, warning alerts are triggered.
- If the viewer continues to stay close, the TV is automatically turned OFF.

The GSM module is tested by verifying that SMS alerts are successfully sent to the registered mobile number.

Fig.2: LED Will GET ON

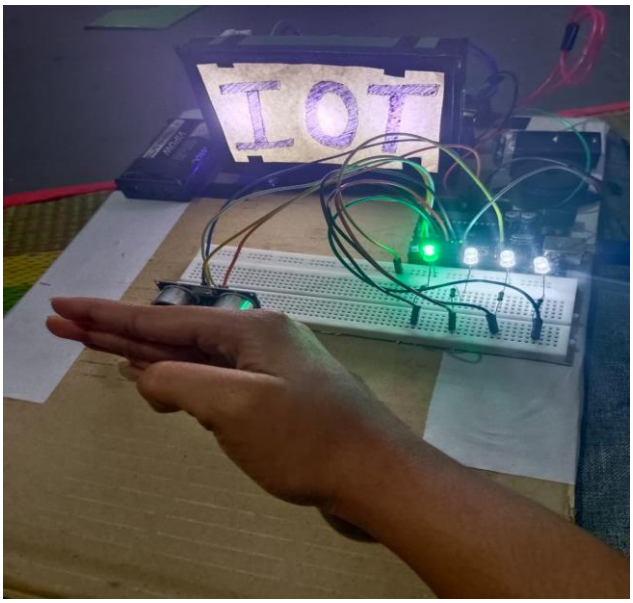


Fig.3: When The Green Light Will Get On (Max Distance Reached By Person).

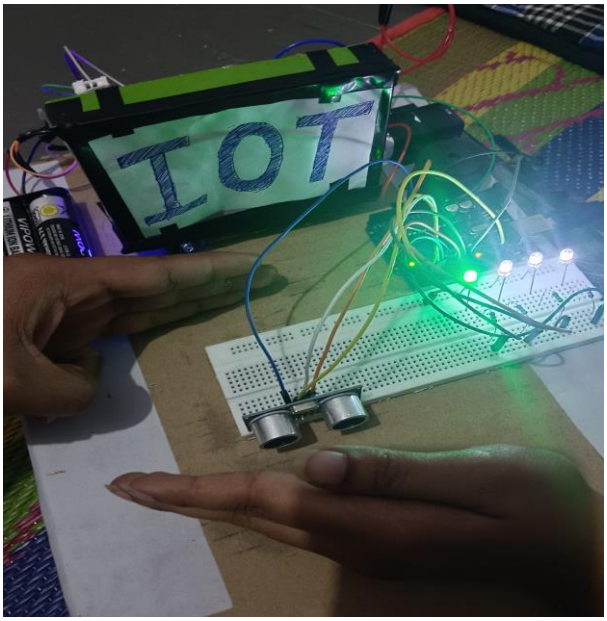


Fig.4: When The Get More Closer To Screen (The Buzzer Will Get On And Screen Will Get Of Automatically)

V. Conclusion & Future Work

In today's digitally driven world, eye care and protection systems have become increasingly important. Continuous exposure to screens can cause eye strain, discomfort, and may even lead to long-term vision problems. Therefore, implementing smart and preventive solutions is essential to protect eye health.

The proposed IoT-based eye protection system provides real-time monitoring and timely alerts to maintain a safe viewing distance and healthy screen usage habits. By measuring distance, tracking user behavior, and generating alerts when necessary, the system helps reduce the risk of digital eye strain. Such smart systems encourage users to follow healthier viewing practices in their daily lives.In addition to technological solutions, preventive measures such as wearing appropriate eyewear, limiting exposure to harmful UV rays, maintaining proper eye hygiene, and taking regular breaks from screen time are equally important. Regular eye examinations also help in early detection and treatment of potential eye-related issues.

The future of IoT-based eye care and protection systems lies in the integration of advanced Artificial Intelligence (AI), Machine Learning (ML), and edge computing technologies to enable intelligent, adaptive, and autonomous health monitoring solutions. Advanced AI-driven algorithms utilizing deep learning, predictive analytics, and behavioral pattern recognition can analyze parameters such as viewing distance, screen exposure duration, blink rate, ambient illumination, and user posture. By leveraging data modeling and real-time signal processing, the system can generate personalized recommendations and adaptive alerts based on historical usage data. The incorporation of reinforcement learning models can further enhance system accuracy by continuously optimizing threshold values according to individual user behavior

Integration with wearable devices such as smart glasses and intelligent contact lenses equipped with embedded sensors, microoptoelectronic components, and biosignal monitoring units will enable continuous ocular

health assessment. These wearables can support real-time feedback mechanisms, blue-light filtering optimization, and automated brightness calibration using ambient light sensors. Cloud-enabled architectures combined with IoT gateways and wireless communication protocols (e.g., Wi-Fi, Bluetooth Low Energy) can facilitate remote data logging, cloud analytics, and longitudinal health tracking. The application of big data analytics and health informatics can support early detection of eye fatigue patterns and potential visual disorders. Furthermore, integration with computer vision techniques using camera-based eye tracking and blink detection algorithms can improve system intelligence and accuracy. The adoption of cyberphysical systems (CPS) architecture ensures seamless interaction between hardware components and software intelligence layers. Overall, the convergence of IoT, AI, wearable technology, and cloud computing will transform traditional eye protection mechanisms into a proactive, intelligent, and scalable digital health ecosystem aimed at long-term visual well-being.

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